

BUCCO

A scientific field guide to his first year

Border Collie · Born Feb 18, 2026 · Home since Apr 27, 2026 · North Hills Pittsburgh

Compiled for Scott Smathers

Spirit of Susan Garrett · Anchored in peer-reviewed science · May 2026

Where Bucco is right now

Week 12 • The cornerstone of the next year

12-18

lb

expected weight range

~880

kcal/day

4 meals • ~80% via training

18-20

hr sleep

REM ≈ 30% of total

16w

window closes

primary socialization

The 12-week brain runs different software than the 12-month brain.

Everything in the next 4 weeks compounds for the next 12 months.

The Four R's

Susan Garrett's foundation priorities · the lens for every decision



Relationship

You are the most reinforcing thing in his world. Built daily, never commanded.



Recall

Comes when called, every time, regardless of distraction. Built in layers, never tested before it's ready.



Retrieve

Brings value back to the handler. The neurochemistry of engagement, not a circus trick.



Reinforcement Zone

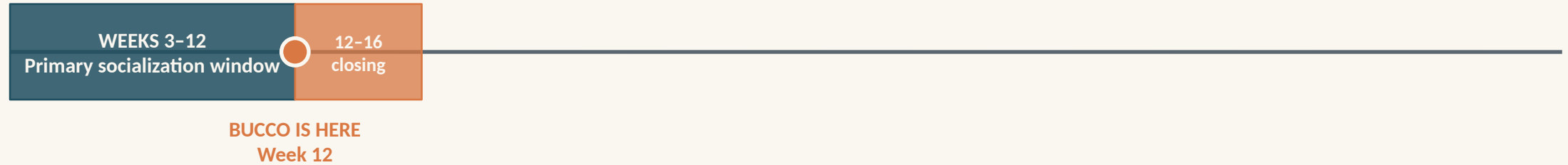
The physical pocket at your chosen side. Food delivered there. Loose-leash, heel, recall — all built on top.

The socialization window

Scott & Fuller, 1965 • Modern review 2022

WEEK 3

WEEK 52



65%

of adult behavior is environment, not genetics

Scott & Fuller, 1965

100

positive new exposures recommended by week
16

AVMA consensus

4 wks

Bucco has left in the primary window

Today: May 12, 2026

Sleep is when learning happens

Puppies don't need exercise. They need rest.

18-20

hours per day, at 12 weeks

REM sleep $\approx$ 30% of total

— vs. $\sim$ 10% in adult dogs.
Iotchev et al., Nature Sci Reports 2021

What this means

- **Hyperactive puppy = under-slept puppy.**
 - Cure is more crate time, not more walks.
- **REM = memory consolidation.**
 - Your 5 min of Crate Games gets filed away while he sleeps.
- **Brain doubles in weight wk 3-12.**
 - Synaptic density peaks then prunes through year 1.
- **Pre-frontal cortex isn't done myelinating until 18-24 months.**

Feeding • the real math

RER × life-stage multiplier • adjust by body condition every 2 weeks

Daily kcal = 70 × (kg)^{0.75} × life-stage multiplier

0–4 mo: ×3.0 • 4–12 mo: ×2.0 • Adult: ×1.4–1.6

At Bucco's current age (12 weeks):

WEIGHT	RER	DAILY (×3)	MEALS
12 lb (5.4 kg)	248	745 kcal	4 ×/day
15 lb (6.8 kg)	294	880 kcal	4 ×/day
18 lb (8.2 kg)	338	1015 kcal	4 ×/day

~80% delivered through training, not from a bowl. Body condition matters more than the scale.

The single most important number on the bag

Hazewinkel et al., Utrecht 1985–1991 · still the standard

Calcium : Phosphorus

1.1 : 1 to 1.3 : 1

for medium and large breed growth

NEVER

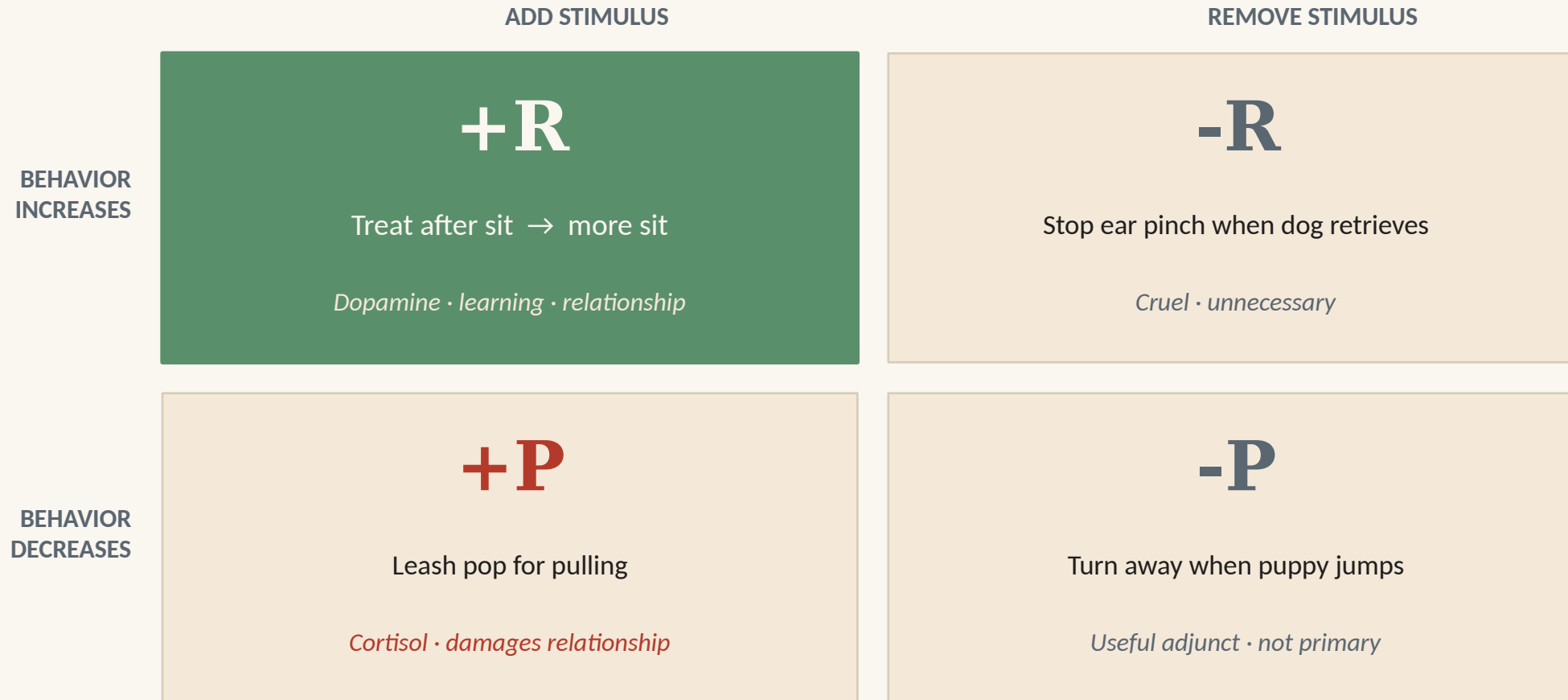
- Supplement calcium 'for stronger bones'
- Feed adult food during growth
- Switch foods every month

ALWAYS

- Check Ca:P on the bag for ratio + values
- AAFCO 'complete & balanced for growth' statement
- Transition new foods over 7–10 days

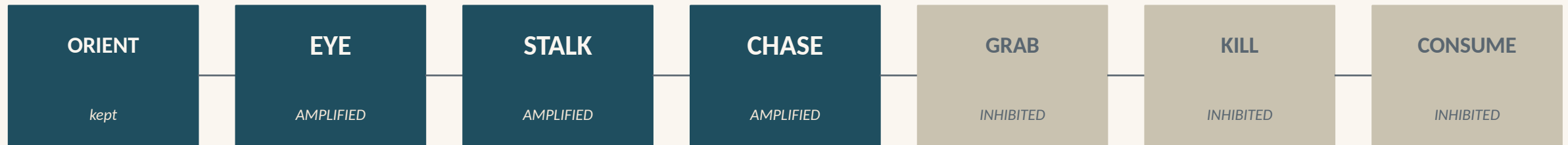
Why we only use the green quadrant

B.F. Skinner 1938 · Modern animal learning science



What makes a Border Collie a Border Collie

Coppinger & Coppinger · the predatory motor pattern, modified



The neural reward sits in the eye-stalk-chase. Not the kill.

Every staring, freezing, slow-stalking moment is its own dopamine pulse. Bucco's brain runs this circuit constantly. Give him legitimate outlets (disc, agility, treibball, nosework) or it invents illegitimate ones.

THE NEVER LIST

Things that, once done, are hard or impossible to undo

Laser pointers

Not once. The BC brain is the most vulnerable canine brain to light fixation. Single sessions have produced lifetime compulsions.

Dog parks

BCs get bullied or develop herding/control problems toward other dogs. Most BC reactivity stories start at a dog park.

Adult food during growth

Wrong calcium/phosphorus profile for medium-breed bones. Orthopedic problems at year 3.

Calcium / vitamin D / vitamin A supplements

All three cause harm in excess. The food has what he needs; more is not better.

Cooked bones

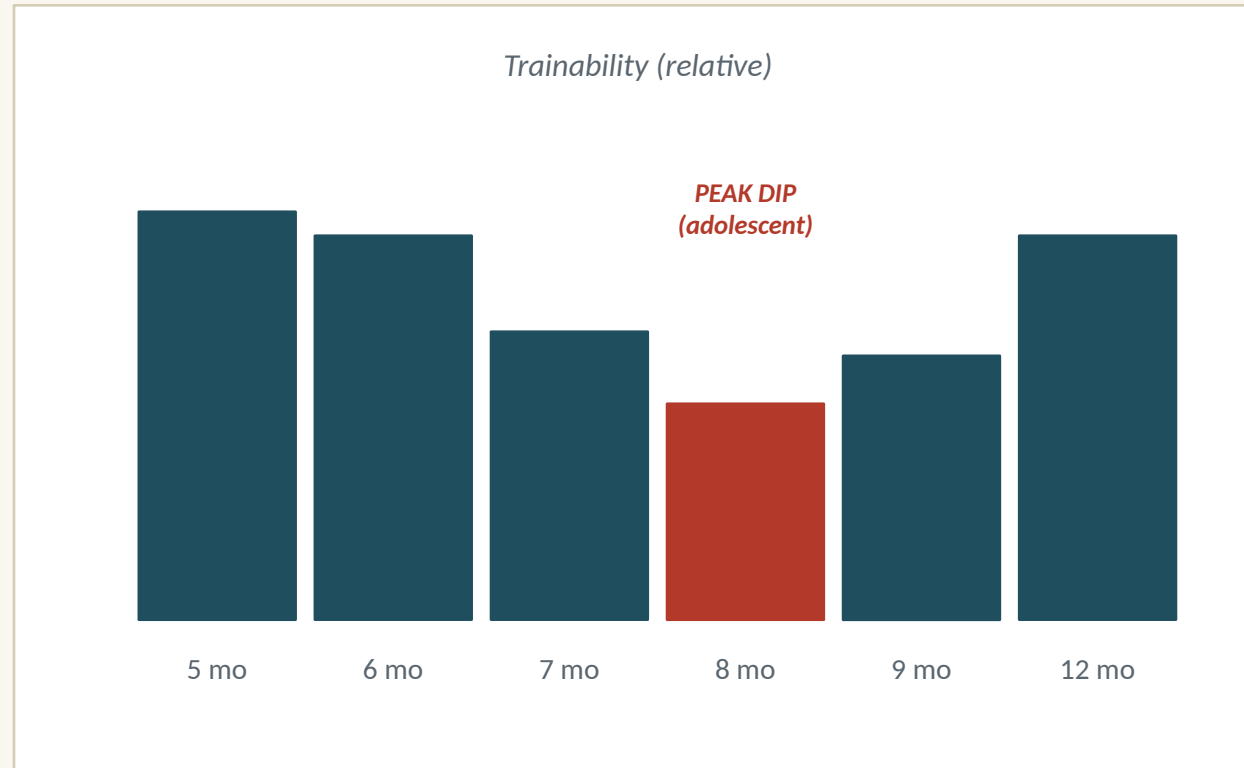
They splinter. Perforate the esophagus, stomach, or intestine.

Sudden food changes

Always transition over 7–10 days. The microbiome needs time.

The 8-month trough is real

Asher et al., Royal Society Biology Letters 2020



What this looks like

- **Trainability dips with YOU, not strangers**
 - Socially specific — he's testing the relationship
- **Hormones + asymmetric brain development**
 - Amygdala fully online; prefrontal still wiring
- **Second fear period may overlap**
- **Maintain, don't expand**
 - Hold ground on what works. New skills wait.

The 30-day plan

May 11 — June 11 • Three skills, deep work

1

Crate Games

The crate is the best place in the world. Door opens = stay. Release cue = freedom. The foundation of every cued behavior.

2

Reinforcement Zone

Pick a side and commit. Food delivered into the zone, not from your hand to his face. Foundation of loose-leash, heel, recall.

3

Restrained Recall

Nina holds. Scott runs and calls. Nina releases. Massive party at the finish. The emotional engine that powers recall later.

Plus: name recognition • It's Yer Choice • cooperative-care pairing for the June 11 vet visit.

When to get professional help

Early intervention works. Late intervention is hard.

Medical / sudden behavior change	VETERINARIAN
Mild, recent behavioral issue	KPA-CTP or CPDT-KA trainer (+R only)
Resource guarding, reactivity	Certified trainer NOW — don't wait
Compulsion (light, shadow, tail)	VETERINARY BEHAVIORIST (DACVB) immediately
Bite incident with intent	VETERINARY BEHAVIORIST + document everything
Severe noise phobia	Vet for meds + DACVB or trainer for desensitization

Cheap on the front end. Expensive on the back end. The math is unambiguous.

What Bucco's capable of

Chaser, the Border Collie who knew 1,022 nouns

1,022

proper nouns learned

Pilley & Reid, Behavioural Processes 2011

Training stopped at 1,022 because Pilley ran out of time. Not because Chaser did.

What this means for Bucco

- The ceiling on what he can learn isn't brain limit — it's your teaching-time limit.
- Words are real cues. He's tracking them whether you mean to teach them or not.
- Trick training, named objects, scent work — these aren't extras. They're his sustainable enrichment.
- 20 min of mental work $\approx$ 2 hr of walking, for tiredness and behavior.

The one sentence

Be more interesting than the environment.

Reinforce the behavior you want.

**Manage the environment to prevent rehearsal of
what you don't.**

Remember that what you grow is what you get.

Everything else is detail.